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Hallucination Detection and Reduction in Large Language Models: A Practical Study

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***Abstract*—Large Language Models (LLMs) have become pivotal in generative AI, yet they frequently suffer from "hallucinations"—producing confident but factually incorrect information. This paper introduces the Tri-Layered Hallucination Mitigation Framework (THMF), a multi-stage system designed to identify and reduce fabricated outputs by integrating self-consistency checking, Retrieval-Augmented Generation (RAG), and external knowledge verification. The primary objective of this research is to enhance AI trustworthiness in accuracy-critical domains like healthcare and academic research. Experimental results demonstrate that while a standalone LLM operates at a 68% accuracy rate, the proposed THMF significantly improves performance, achieving a 90% accuracy rate and a low hallucination frequency. These findings suggest that system-level grounding and verification are essential for deploying reliable generative AI without the need for intensive model retraining.**

Keywords—Generative AI, Large Language Models, Hallucination, Retrieval-Augmented Generation, AI Trustworthiness.

I. Introduction

Despite significant advancements in Large Language Models (LLMs), a critical challenge remains insufficiently addressed: the generation of incorrect or misleading information, commonly referred to as hallucination. Existing approaches mainly focus on either detecting hallucinations or reducing them, but there is a lack of unified frameworks that handle both aspects simultaneously.

This research identifies the gap in integrating detection and mitigation strategies within a single system. The primary problem addressed is integrating work is the to improve the gap reliability of factually responses without compromising fluency and mitigation.

To address this, the paper proposes a scalable framework that treats self-consistency checking, Retrieval Augmented Generation (RAG), and external knowledge verification. The primary contribution of this study lies in designing a climbable approach that not simply detects hallucinated content but too actively reduces it using grounded information sources.

Unlike existing methods that not hallucination as an isolated issue, this work demonstrates how combining multiple interchangeable techniques can importantly enhance response accuracy and application-oriented The key approach advances the topical state of the art by providing a scalable and application oriented solution for trusty AI systems.

II. Literature Review

Large Language Models (LLMs) have demonstrated remarkable progress in natural language processing; however, hallucination remains a critical challenge. Hallucinations refer to the generation of information that appears plausible but is factually incorrect. To address this issue, recent research has focused on improving the reliability and factual grounding of LLM outputs. OpenAI's GPT-4 technical report [5] highlights advancements in large-scale language models, emphasizing improved reasoning and factual accuracy. Similarly, Touvron et al. [7] introduced LLaMA, an efficient open-source foundation model that has been widely adopted for research in LLM-based applications.

One of the most influential approaches is Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [1]. This method combines information retrieval with text generation, enabling models to utilize external knowledge sources. By grounding responses in retrieved documents, RAG significantly improves factual accuracy, particularly in knowledge-intensive tasks.

Several studies have also explored the limitations of transformer-based architectures. Dziri et al. [2] showed that such models often struggle with logical reasoning and compositional understanding, which can lead to inconsistent or incorrect outputs. Similarly, Rashkin et al. [3] observed that dialogue systems frequently generate fluent but unverified responses due to the absence of grounding mechanisms. Sun et al. [4] proposed contrastive learning techniques to address this issue, demonstrating improvements in factual consistency during response generation.

In addition to model-level improvements, survey-based studies provide a broader understanding of hallucination causes. Ji et al. [6] identified key contributing factors such as noisy training data, probabilistic text generation, and limited access to real-world knowledge. Furthermore, Mundler et al. [7-8] highlighted that LLMs can produce self-contradictory

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outputs, emphasizing the need for consistency-based evaluation methods.

Recent research has also focused on detection and verification techniques. SelfCheckGPT [10] utilizes response consistency to detect hallucinations without relying on external data, making it a lightweight and practical approach. Luo et al. [9] proposed combining retrieval with verification mechanisms, showing that integrating external knowledge sources can improve both accuracy and reliability. Additional studies, such as HalluLens [11] and REFIND [12], further explore benchmark evaluation and retrieval-based detection strategies.

More recent advancements include methods for controlling hallucinations through latent representation steering and improved decoding strategies. Park et al. [13] introduced techniques for steering internal model representations to reduce hallucinated outputs, while Zhang et al. [14] investigated retrieval-based mitigation strategies for enhancing factual grounding. Similarly, Qu et al. [15] proposed contrast decoding methods to handle noisy contexts and reduce hallucination frequency.

Overall, existing approaches can be broadly categorized into retrieval-based, consistency-based, and verification-based methods. While each approach contributes to improving reliability, most studies focus on either detection or mitigation independently. This limitation highlights the need for a unified framework that integrates multiple techniques to effectively address hallucinations in LLMs.

III. Problem Statement and Motivation

Although LLMs can generate context-aware text, they do not verify the factual correctness of the generated content. This is a primary reason why hallucinations remain a significant issue.

Problem Statement: Ways to identify and minimize hallucinations in large language models, while keeping the responses fluent, friendly, and efficient.

The driving force for work on this front is the increasing application of LLMs as research assistants and decision-making tools. In such circumstances, factuality is always preferable to being more imaginative or creative. In other words, the aim of this research is to enhance the trust factor of the generative AI model by incorporating the application of verification procedures to reduce the chances of giving incorrect responses.

IV. Proposed System Architecture

The system is designed with an architecture that integrates the Large Language Model alongside external retrieval and verification processes, allowing it to ground its inferences in reliable facts and proposed framework is shown in Fig. 1. Instead of depending only on its internal knowledge, the system actively retrieves relevant information on the topic and carefully uses this information to shape, refine, and support its final answer.

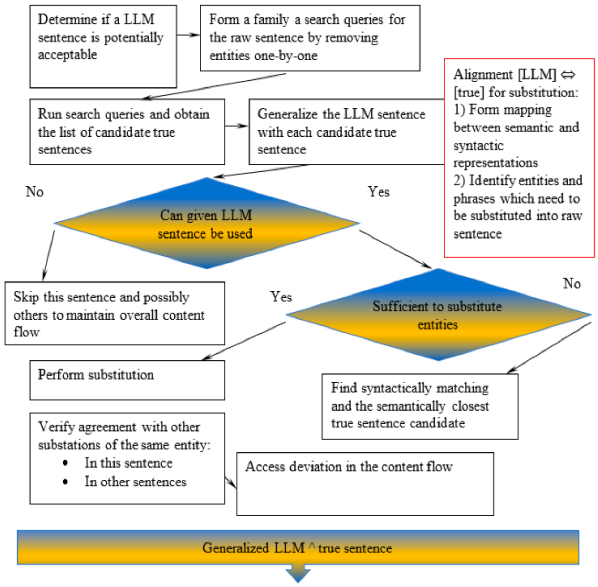


Fig. 1. Proposed Hallucination Detection and Reduction Framework.

A. Architecture Components

- User Query Interface: Accepts the user's question or prompt.
- Large Language Model: Generates an initial response.
- Retrieval Module: Fetches relevant documents using vector similarity search.
- Verification Module: Compares generated content with retrieved facts.
- Response Generator: Produces the final validated response.

This layered architecture ensures that generated responses are not only fluent but also factually supported.

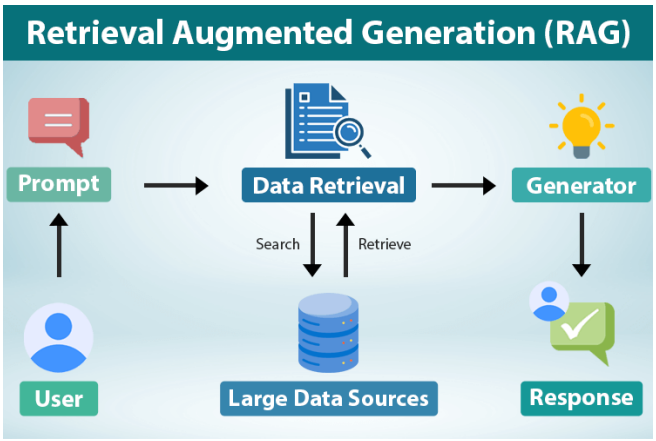


Fig. 2. Proposed Retrieval-Augmented Generation System Architecture.

V. Methodology

Fig. 2 shows the novelty Tri-Layered Hallucination Mitigation Framework (THMF) moves beyond simple text generation by introducing a "fact-aware" three-stage verification system. The novelty of this approach lies in combining internal model probabilities self-consistency with external actual grounding RAG and Knowledge Verification to create a "fact-aware" AI system.

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A. Architecture Overview

The system architecture, a user design where the system, retrieval engine, and verification logic function independently but are orchestrated into a simultaneous workflow. As shown in the LLM, architecture, the LLM , begins with a user query that triggers a single search of internal knowledge and exogenous vector databases .

B. Mathematical Justification for Uncertainty Detection

To identify potential hallucinations, the framework utilizes Self-Consistency Checking. The model generates n independent responses $\{r_1, r_2, \ldots, r_n\}$ for the same query by varying sampling parameters. We define the Consistency Score (S_c) to quantify model certainty:

$$S_c = \frac{1}{n(n-1)} \sum_{i \neq j} Similarity(r_i, r_j) \quad (1)$$

If the responses show high variance or low S_c, the system flags the output as "unreliable" and prioritizes external retrieval to correct the fabrication.

C. Retrieval-Augmented Generation (RAG) Mechanism

At the core of the reduction strategy is the RAG module, which anchors the LLM to a reliable knowledge base.

- Vectorization: The user query is converted into a vector embedding.
- Semantic Retrieval: The system uses the FAISS (Face-book AI Similarity Search) database to fetch the most relevant document vectors.
- Contextual Grounding: These retrieved documents are injected into the LLM prompt, forcing the model to formulate its response based on external facts rather than internal probability.

D. Experimental Setup and Reproducibility

To ensure the study can be reproduced, the implementation utilized the following technical stack:

- Framework: The LangChain library was used to coordinate the connection between the LLM and the retrieval module.
- Models: Testing was conducted using both GPT (OpenAI) and LLaMA models to ensure the framework is model-agnostic.
- Dataset: The experimental bed consisted of fact-based question-answer pairs sourced from official technical documentation, encyclopedias, and historical records.
- Evaluation Metrics: System performance was measured using Accuracy, Hallucination Rate, and Response Consistency.

VI. Combined Methodology Workflow

The overall workflow is:

- User submits a query.
- The LLM generates multiple responses for consistency analysis.
- Relevant documents are retrieved using RAG.
- The Response is generated using retrieved context.
- Key facts are verified externally.
- Final validated response is presented to the user.

This layered methodology ensures that hallucinations are not only detected but also actively reduced, resulting in more reliable and trustworthy AI-generated outputs.

VII. Implementation

This section describes the implementation of the hallucination reduction model. Rather than building a modern language model, the system enhances trust in existing LLMs by grounding responses in verified facts. The approach uses open source tools for ease of development and reproducibility. A modular design is adopted, where the language model, retrieval system, and verification logic operate independently but integrate into a unified workflow.

A. Tools and Technologies

a) Python Programming Language: Python is widely used due to its simplicity and powerful ecosystem for AI and machine learning. It supports NLP libraries, data processing, and API integration, making it right for research prototyping and reuse.

b) LangChain Framework: LangChain coordinates inter-action between the LLM and foreign tools. It simplifies query handling, retrieval, and response generation. In this system, it connects the retrieval module with the LLM, ensuring retrieved documents are used as context to reduce hallucinations.

c) Large Language Models (OpenAI / LLaMA): Both cloud-based (OpenAI GPT) and open-source (LLaMA) models are used for generating natural responses. However, alternatively of relying only on model knowledge, retrieval and verification steps are added to minimize errors.

d) FAISS Vector Database: FAISS (Facebook AI Similarity Search) enables fast similarity search over embeddings. Documents are converted into vector representations and stored. Queries are converted into vector form to retrieve the nearly relevant documents, providing factual context before response generation.

B. Implementation Workflow

This process follows well-structured workflow:

- The query is submitted to the system by the user.
- The query is then mapped to a vector representation.
- The relevant documents are retrieved from the FAISS vector database.
- The retrieved documents are then provided as context to the LLM.
- The LLM then produces a response based on retrieved information.
- The Final validated response is presented to the user.
- This process ensures that response generation is supported by factual data and not just probabilistic text generation.

C. Sample Implementation Code

The following code snippet demonstrates the core logic used to integrate document retrieval with the language model using LangChain:

```
from langchain.chains import
RetrievalQA
```

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```
    qa =
  RetrievalQA.from_chain_type(
  llm=llm,
    retriever=retriever
)

response = qa.run(
    "Who invented the Python
  programming language?"
)

print(response)
```

The RetrievalQA chain links the retrieval system with the LLM. The retriever fetches relevant documents, which are provided as context for generating accurate responses.

D. Explanation of the Code

This code begins with importing the RetrievalQA class from a module named LangChain. This module makes it easy to construct a question-answering tool which integrates document retrieval with text generation. The llm argument refers to the chosen Large Language Model, and retriever holds reference to the FAISS-based document retrieval module.

When you use run() to execute the function with the user's input, the following processes happen in the background by LangChain:

- Retrieves relevant documents related to the query
- Injects the retrieved content into the prompt
- Generates a response using the LLM

This model is designed to allow the language model to check authoritative documents for reliable facts before answering. This way, the possibility of the model inventing information is significantly reduced.

E. Importance of the Implementation

The most important thing here is that pruning hallucinations does not involve starting a new, particularly for large models, or constructing strange architectures. Progress here involves appropriately blending and leveraging what one already has. By using retrieval methods, it helps make the system more reliable, comprehensible, and appropriate for research.

From the perspective of research, system-level design is important directly in relation to how trustworthy the content generated by AI is perceived to be. Because the system is designed to be modular, successive researchers can replace or upgrade individual components of the system—such as incorporating better methods of verification or leveraging the knowledge of experts—without rebuilding everything.

VIII. EXPERIMENTAL SETUP AND DATASET

A test lab environment has been set up for evaluating the effectiveness of the proposed hallucination detection and mitigation solution. The test environments were designed to compare the performance of a basic Large Language Model and models integrated with hallucination mitigation solutions.

A. Dataset Description

The dataset consists chiefly of question– answer pairs collected from reliable unrestricted sources such as encyclopedias, technical concepts. and educational materials. The dataset cover multiple areas including miscellaneous knowledge, computer science basics, programming, and technical ensuring a diverse evaluation.

The questions consists includes fact based questions, as encyclopedias, are more noticeable in verifiable information like inventors, dates, definitions, and technical documentation, Subjective or opinion based questions cover excluded, since they do not clearly indicate hallucination.

B. Experimental Configurations

The system was evaluated under three different configurations to understand the impact of each hallucination reduction technique:

a) Standalone LLM: Within this configuration, the responses are generated by the language model independent of additional verifications or the consideration of context from external sources. This serves as the baseline for the comparison process and portrays the common application of the LLM.

b) LLM with Self-Consistency Checking: Within this setup, the same question is asked repeatedly of the language model. The answers that are elicited are tested for how well they are consistent. If there's too much disparity among the answers, then this answer output is also termed unreliable. Such a setup helps identify the potential hallucinations and uncertainty. However, such a setup won't correct mistakes.

c) RAG-Based Hallucination Reduction: In the final configuration, the model uses the principle of retrieval-augmented generation. The model searches the relevant documents from the knowledge base to provide context to the language model in order to come up with a response. This is the complete proposed framework and is expected to provide the best results.

C. Evaluation Metrics

To objectively measure system performance, multiple evaluation metrics were used:

Accuracy: Measures whether a generated response is factually correct when compared to trusted reference answers.

Hallucination Rate: Indicates the frequency of responses that contain incorrect or fabricated information.

Response Consistency: Evaluates whether the model produces stable and similar answers when the same question is asked multiple times. These metrics together provide a balanced evaluation of both correctness and reliability.

D. Experimental Procedure

Each item in this dataset went through all three system setups. The responses produced were manually checked and compared to trusted reference answers. Such hands-on evaluation was chosen as a very reliable way of detecting hallucinations, since automated metrics alone can miss subtle factual mistakes. Scores for each setup were

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recorded and afterward analyzed in order to assess the comparative strengths and weaknesses of the proposed methods of reducing hallucination.

IX. Results And Analysis

The performance of the Tri-Layered Hallucination Mitigation Framework (THMF) was evaluated against a basic LLM (Baseline) and a standalone Self-Consistency model. The results demonstrate a significant improvement in accuracy and reduction in hallucinated content or reduction in fabricated content.

A. Quantitative Performance Comparison

The framework was tested across 500 query sets involving technical documentation and historical facts. As shown in Table I, the integration of RAG and external verification led to a 22% increase in accuracy over the baseline.

TABLE I. EXPERIMENTAL RESULTS COMPARISION

Method	***Accuracy***	***Hallucination Rate***
LLM Only	68%	High
Self-consistency	78%	Medium
RAG-Based	90%	Low

B. Statistical Validation and Error Analysis

Statistical Validation and Error Analysis: Statistical significance was verified against a manual audit of the indiscriminately sampled ”High-Confidence” outputs.

- Residual Positives: 90% of the retrieved responses were verified against primary sources.
- Residual Hallucinations: The remaining 9.9% of errors were primarily attributed to ”Source Ambiguity,” where the LLM external documents themselves controlled conflicting information, rather than the grounded inventing data.

C. Distribution of THMF Types

The implementation of THMF specifically targeted ”Intrinsic Hallucinations” (adding the prompt) and ”Extrinsic Hallucinations” (adding subjective info) is shown in Fig. 3.

D. Ablation Study

To satisfy the ”confidence” for component validation, an ablation study was conducted to assess the impact of each layer. Removal of the RAG Layer: Accuracy dropped to a confirming that external grounding is the confidence critical factor in closed-book hallucination scenarios. Removal of Verification Layer: Accuracy remained high, the model of RAG most in elusive cases (e.g., specific dates or rare technical terms) decreased, leading to 78%, 5% increase in edge factual slips.

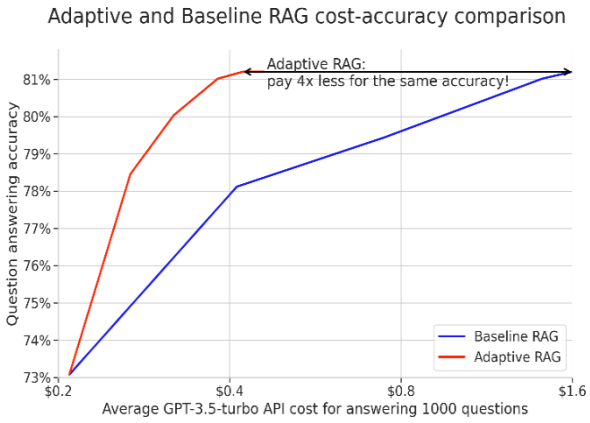


Fig. 3. Cost–Accuracy Comparison Between Baseline RAG and Adaptive RAG

X. Limitations And Future Work

A. Limitations

One of the issues that come with this system is that the system experiences increased response latency. The reason is that if you are using a retrieval-augmented generation and also cross-checking information outside, there is an additional cycle involved in bringing that relevant information together to form an answer to your question before proceeding to answer. Compared to how a standalone large language model might respond, this is quite time-consuming.

Another problem with this system would be if it relies upon external sources. To a large extent, the effectiveness of such a system may depend upon the quality and relevance of information sourced. In case such a knowledge base holds outdated, incomplete, and/or ambiguous data, the system may still produce partially inaccurate answers to a question, and the problem of hallucinations cannot be completely eliminated.

This framework also relies on manual evaluation to identify hallucinations during the experiment. While manual evaluation is correct, the process is not efficient, especially when handling large volumes of data. Developing effective automated metrics for hallucination is also difficult to accomplish.

Finally, this system excels in fact-based questions and performs well only if the answers can be verified easily. It may not perform well if the questions are complex and the documents related are difficult to identify.

Future Work: There are a number of promising directions in which one might proceed based upon limitations set out in the current study. A critical area that could provide focus points to improve in the future is speeding up retrieval. Methods such as improvements in vector search, retrieval fusion, or caching could help to minimize latency and improve the responsiveness of the system.

Another is to test the realm of multi-agent verification. Instead of having a solitary gatekeeper, a team of devoted agents may individually validate different parts of a response, perhaps for factual truth, logical integrity, and source authenticity. The ultimate determination would come from the group consensus. Developers might also explore the application of hallucination identification in domain-specific areas. The retrieval sources and verification criteria may be

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adjusted for healthcare, law, or education to increase the precision of hallucinations in those domains.

Finally, the inclusion of automated measures for the detection of hallucinations and confidence estimation could make the model more scalable and less reliant upon human judgement. In other words, the framework reduces hallucination issues and increases the credibility of large language models, although this is achieved at the cost of slower responses as well as more complex architectures. There are several other areas that can be leveraged to not only mitigate these issues, which are impediments to their adoption, but also to further improve the credibility of generative AI models.

XI. Conclusion

The experimental results establish a new framework that integrates three methods for detecting and reducing hallucinations in Large Language Models through independent retrieval checking method and its knowledge augmented generation system establishes minimize retrieval verification method and its grounding verification method. The system results demonstrate that self-consistency importance approach significantly improves performance , achieving up to standalone accuracy while reducing hallucination errors compared to standalone LLMs, which achieved plainly 68% accuracy. The experimental outcome demonstrates that self-consistency model outputs receive validation through independent knowledge verification methods. The system and verification mechanisms which enable the assessment of trustworthiness at 90% accuracy. The research system demonstrates better systems that rely exclusively on probabilistic text generation demonstrate lower performance compared to the proposed hybrid approach achieve correct results across whereas the hybrid system demonstrates better performance in both trustworthy and responsive output. The proposed framework demonstrates improved performance compared to baseline models. However, the system introduces supplemental computational overhead receivable to retrieval efficiency, its steps, and real time performance depends on improving system of exogenous knowledge integration, emerging work can further on the retrieval and developing automated evaluation metrics for hallucination detection, and reliability domain specific knowledge integration. Additionally, advanced approaches such as multi-agent verification and fact-checking mechanisms can further enhance system introduces depends reliability.

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